HW3

1)

```
 1011   → 11
 1101   → 13
11|1000 → 24 ≠ 8
```

overflow, can't represent using 4 bits, for unsigned addition.

2).

```
 1011   → -5
 1101   → -3
 11000  → -8
```

Since this is signed, -8 can be represented using 4 bits, no overflow.

3)

Multiplicand	Multiplier	Product	Description
00000 11010	11101	00000 11010	H(0)=1, Add multiplicand to product
00001 10100	01110	00000 11010	Shift both.
00011 01000	00111	00000 11010	Shift both
00011 01000	00111	00100 00010	Add multiplicand
00110 10000	00011	00100 00010	Shift both
00110 10000	00011	01010 10010	Add multiplicand
01101 00000	00001	01010 10010	Shift both
01101 00000	00001	10111 10010	Add multiplicand
11010 00000	00000	10111 10010	Shift.

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Multiplier 11101.

4.

Multiplicand	Product	Description
11010	00000 11101	Add multiplicand to product
11010	11010 11101	Add multiplicand to MSB
11010	01101 01110	shift product
11010	00110 10111	shift »
11010	+ 00000 10111	Add multiplicand to MSB
11010	10000 01011	shift
11010	+ 01010 01011	Add
11010	10101 00101	shift
11010	+ 01111 00101	Add
11010	10111 10010	shift

5). $-15.25_{10} = -(01111.01_2)$

$= (-1)^1 \times (1.11101_2) \times 2^3$.

here $S = 1$,

fraction $= 11010\ldots0_2$

exp $= 3 + 127 = 130_{10}$. (∵ exp-bias=3)

Bias

Single precision	1	$1000\,0010_2$	$11010\ldots0_2$
Double precision	1	$100\,0000\,0010_2$	$11010\ldots0_2$

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6). Given: 129

0 | 1000 0001 | 10101110...0

$= (-1)^0 \times 1.10101110_2 \times 2^{?}$

exp = 129 − 127 = 2.

128

0 | 1000 0000 | 10101010...0

$= (-1) \times 1.10101010_2 \times 2^{?}$

exp = 128 − 127 = 1

⇒ difference in exponents = 2 − 1 = 1.

Shift the number with the smaller exponent by the difference.

0.110101010×2^2.

Add the numbers

$$\begin{array}{r} 1.10101110 \\ +\ 0.11010101 \\ \hline \rightarrow 10.10000011 \end{array}$$

Normalizing → $1.0100000110_2 \times 2^3$.

In single precision

0 | 1000 0010 | 01000 00110...0

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